

Biliary POCUS — Cholelithiasis & Cholecystitis Quick Reference

RUQ ultrasound: gallbladder anatomy, stone vs sludge, wall thickening, cholecystitis. TeachIM 5-minute POCUS case · CC BY-NC 4.0 · clinician-reviewed.

One line: A gallstone is gravity-dependent echogenic material with a hyperechoic rim and a posterior **acoustic shadow** — the shadow is what separates a stone from sludge or a polyp. Wall thickening is suggestive but non-specific.

Technique

- **Probe:** low-frequency **curvilinear** (abdominal).
- Image in **long** AND **short** axis of the GB.
- **Fasting** GB distends & is easier to see; non-fasting = contracted + bowel gas.
- Landmarks: liver, GB, portal triad / main lobar fissure.

Stone vs Sludge vs Polyp

	Echogenic?	Gravity-dep?	Shadow?
Stone	Yes (rim)	Yes	Yes
Sludge	Yes	Yes (layers)	No
Polyp	Yes	No (fixed)	No

- **Acoustic shadow = stone.** Tumefactive sludge can mimic a mass but casts NO shadow.

Wall Thickening

- Normal GB wall **<3 mm** (TG18 dx threshold **>4 mm**).
- Non-specific — also seen in **cirrhosis, hepatitis, pancreatitis, hypoalbuminemia**.
- Other cholecystitis signs: pericholecystic fluid, GB enlargement, impacted neck stone.

Cholecystitis — TG18 Criteria

A — Local	Murphy's sign; RUQ mass/pain/tenderness
B — Systemic	Fever; ↑CRP; ↑WBC
C — Imaging	Wall >4 mm, pericholecystic fluid, stones/debris, enlarged GB (LA ≥8 / SA ≥4 cm), sonographic Murphy

- **Definite dx = one A + one B + one C.**
- **Sonographic Murphy** (max tenderness under the probe) = key POCUS sign.

When to Worry About the Duct

- **Normal LFTs** → choledocholithiasis & cholangitis unlikely.
- CBD imaging is technically hard & low-yield with normal LFTs.
- **Abnormal LFTs** → pursue CBD / MRCP; consider cholangitis (Charcot/Reynolds).

Bedside Bottom Line

- RUQ pain → curvilinear probe, find the GB in 2 planes.
- Echogenic + **shadow** = stone; look at the GB **neck**.
- Wall >3–4 mm + sonographic Murphy + systemic signs → cholecystitis.
- Check LFTs before worrying about the duct.

Sources: Soni NJ, Point-of-Care Ultrasound, 2nd ed (Elsevier 2019); Yokoe M et al, Tokyo Guidelines 2018 (TG18), J Hepatobiliary Pancreat Sci 2018; case image Radiopaedia rID 19516 (O'Donnell). Adapted from TeachIM.org "Cholecystitis – POCUS" (Doran, Fainstad; ed. Ortiz-Lopez), CC BY-NC 4.0. Educational use only.